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Reflection Journal

Brian: Introduction + Linear Regression Code + Knowledge of the Workflow

In this lab, I worked on the introduction and the linear regression part of the notebook using the Titanic dataset. I helped write the code, added comments to explain the main steps, and ran the code to check the outputs. At first, I found it a little difficult to understand how data preparation, model training, prediction, and evaluation were connected. After going through the code, I started to understand how each step works together in the machine learning process.

For the linear regression section, I worked with a model that predicts passenger fares using age and passenger class. I used LinearRegression() to create the model, .fit() to train it, and .predict() to make predictions. I also learned about Mean Squared Error (MSE), which measures the difference between predicted fares and actual values. This helped me understand why we need to evaluate a model instead of only checking whether it can make predictions.

Toan: Linear Regression + Code Explanation + Improvements and Learning

My main focus in this lab was the linear regression section. I worked on the code that used passenger age and class to predict ticket fares. This included creating a LinearRegression()

model, training it with .fit(), and using .predict() to estimate fares from the test data. I also added comments to explain the important steps and make the code easier to read.

One important part of this section was evaluating the model with Mean Squared Error (MSE). I learned that MSE compares predicted fares with the actual values and helps show how much error the model makes. A lower MSE generally means the predictions are closer to the actual values. This helped me understand why we need to check the results after training a model.

During this lab, I improved my ability to follow Python code and understand how different functions are used in a machine learning project. Sometimes, I needed to review a section more than once to understand how the steps worked together. In future assignments, I want to become more confident writing code and explaining the results without relying too much on examples. I can also use what I learned here when working with other datasets and regression models.

Montserrat: Logistic Regression + Code Explanation + Improvements and Learning

The part I focused on was logistic regression, which was used to predict whether Titanic passengers survived. The model used information such as age, passenger class, and sex. I worked through the code for creating and training the model with LogisticRegression() and .fit(), then used .predict() to get the predicted survival outcomes. I also worked with the step that converts the sex column into numerical values so it can be used by the model.

What stood out to me was the difference between regression and classification. Linear regression predicts numerical values, such as passenger fares, while logistic regression predicts categories, such as whether someone survived. I also looked at accuracy, which measures how many predictions matched the actual outcomes. This showed me why the evaluation method needs to match the type of problem being solved.

One thing I improved during this lab was my understanding of how classification models use data to make predictions. I also became more familiar with checking model accuracy and reviewing the outputs. For future labs, I want to practice more with different machine learning models and become more comfortable working with Python. I can apply these skills when completing other assignments that involve preparing data, training models, and evaluating predictions.

Conclusion

Overall, this lab helped me understand more about supervised machine learning and how models are trained and evaluated. I learned how linear regression can predict numerical values and how logistic regression can be used for classification. I also became more familiar with MSE and accuracy and how they are used to check model performance. Working with the Titanic dataset gave me practical experience with these concepts, and I can build on this experience in future machine learning assignments.